



**MBARARA UNIVERSITY OF SCIENCE AND
TECHNOLOGY.**

FACULTY OF APPLIED SCIENCES AND TECHNOLOGY

TITLE: DEVELOPMENT OF A LOW COST THERMAL IMAGING DEVICE FOR
SCREENING BREAST CANCER.

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A Project Proposal Submitted to the Faculty of Applied Sciences and Technology for the Study Leading to a Project in Partial Fulfillment of the Requirements for the Award of the Degree of Bachelor of Biomedical Engineering of Mbarara University of Science and Technology.

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October 2025

Chapter 1: Introduction

1.1 Background

Breast cancer is a disease characterized by uncontrolled growth of epithelial cells within the breast ducts or lobules caused by genetic, hormonal, and environmental factors [1]. The disease exists in four stages with the first two treatable if diagnosed early. It is the most commonly diagnosed cancer and late diagnosis remains one of the leading causes of deaths among women worldwide [2]. Despite advances in treatment and awareness, the disease continues to pose a major global health burden, particularly in low- and middle-income regions where access to early detection and specialized care is limited. In Sub-Saharan Africa (SSA), women diagnosed with breast cancer are up to four times more likely to die from the disease than those in high-income countries, primarily due to limited screening infrastructure, financial constraints and social stigmas leading to late diagnosis [3], [4].

The current breast cancer diagnostic techniques include X-ray mammography, breast MRI, ultrasound, CA15-3 and physical examination but most of these techniques are expensive, require specialized expertise, and may expose patients to ionizing radiation [5]. In many SSA countries, such infrastructural and economic limitations mean that breast cancer is typically detected at advanced stages. This situation emphasizes the urgent need for a non-ionizing, cost-effective, and widely accessible diagnostic tool that can be used to screen breast cancer.

Thermal imaging, also known as Infrared Thermography (IRT), has emerged as a promising functional imaging technique to address these gaps [6]. Unlike the other existing diagnostic techniques which visualizes structural features, IRT detects physiological changes by capturing thermal radiation emitted from the skin surface. Cancerous tissues generate localized heat due to increased metabolic activity, angiogenesis, and abnormal blood flow, creating temperature patterns in the form of infrared radiation that can indicate malignancy before structural changes

become visible [7]. IRT is non-contact, radiation-free, painless, and highly portable, making it ideal for deployment in low-resource settings.

1.2 Problem Statement

Breast cancer remains one of the leading causes of cancer-related deaths among women worldwide, accounting for approximately 685,000 deaths annually according to the World Health Organization (2023) [8]. It represents one in four of all cancer cases [9] and one in six of all cancer deaths among women [10]. The disease burden is particularly severe in low- and middle-income countries (LMICs) where access to timely screening and diagnostic services is limited. In Sub-Saharan Africa (SSA), for instance, more than 70% of breast cancer patients are diagnosed at advanced stages when treatment outcomes are poor [11]. These high mortality rates emphasize the urgent need for early, accessible, and effective breast cancer detection methods but most of the current diagnostic methods which include mammography, PET, breast MRI, ultrasound and physical examination are limited by low sensitivity, high cost, low specificity, radiation exposure and stigma all of which hinder efficient diagnosis [12].

Thermal Imaging (Infrared Thermography, IRT) presents a promising functional imaging solution by detecting the localized, subtle heat signatures generated by increased metabolic activity in malignant tumors [13]. Although IRT is non-invasive, radiation-free, and highly portable, its clinical utility as a standalone screening tool is currently limited by the need for standardized imaging protocols and the robust validation of automated interpretation systems capable of translating thermal patterns into reliable diagnostic conclusions, especially within the challenging healthcare landscape of Sub-Saharan Africa [14].

1.3 Objectives

1.3.1 Main Objective

To improve early breast cancer screening in Low and Middle Income countries by developing a low-cost thermal imaging device.

1.3.2 Specific Objectives

1. To review and understand the available medical thermal imaging technologies and identify possible future research directions.

2. To develop an infrared based device to capture signals from the breast ROI and evaluate its performance.
3. To develop an algorithm for interpreting infrared signals in real time and evaluate it.

1.4 Research Questions

1. What are the limitations of existing breast cancer diagnostic methods, particularly for women with dense breast tissue and in low-resource settings?
2. How can a low-cost thermal imaging device be designed and optimized for effective breast cancer screening?
3. To what extent can thermal imaging, combined with image processing algorithms, accurately detect early-stage breast abnormalities compared to standard screening methods

1.5 Scope of the Study.

This project focuses on the design, development, and evaluation of a low-cost thermal imaging device for early breast cancer screening. It includes a review of existing diagnostic methods to identify their limitations, particularly for women with dense breast tissue and in low-resource settings. The project will involve constructing a functional thermal imaging prototype, developing image processing algorithms to detect abnormal thermal patterns, and evaluating the system's accuracy, sensitivity, and reliability under controlled conditions. While full clinical deployment is beyond the project's scope, the outcome will provide a validated, portable, and affordable screening tool suitable for early-stage detection and future integration into resource-limited healthcare programs.

1.6 Significance and Justification.

Breast cancer remains a leading cause of death among women worldwide, with the highest mortality rates occurring in low- and middle-income countries due to late detection and limited access to effective screening [15]. Conventional methods like X-ray mammography are often less effective for women with dense breasts, expensive, and inaccessible in resource-limited settings,

leaving a significant portion of the population underserved [16]. Developing a low-cost, non-ionizing, and portable thermal imaging device addresses these challenges by providing an affordable, safe, and easily deployable alternative for early detection. This project is significant because it has the potential to reduce diagnostic delays, improve survival rates, and increase accessibility to breast cancer screening, particularly in underserved communities. By leveraging thermal imaging and intelligent image analysis, the project provides a practical, scalable solution that complements existing screening programs and supports equitable healthcare delivery.

Chapter 2: Literature Review.

Breast cancer is a heterogeneous group of malignant diseases originating from the epithelial cells of the breast, most commonly in the ductal or lobular tissues [17]. According to the Encyclopedia Britannica, most cases arise within the ducts (infiltrating ductal carcinoma, about 70%) or the lobules (lobular carcinoma, under 15%), and are further classified by receptor status—such as estrogen-receptor positive, progesterone-receptor positive, HER2-positive, or triple-negative—which influences both tumour behaviour and treatment response [18]. A wide range of factors contribute to breast cancer risk, including inherited mutations (e.g., BRCA1/2), hormonal exposures (early menarche, late menopause, prolonged estrogen/progesterone exposure), reproductive history, lifestyle factors, and breast tissue characteristics such as high density [19], [20]. Dense breast tissue not only increases cancer risk but also complicates detection. While early-stage, receptor-positive cancers are often highly treatable with surgery, radiotherapy, endocrine therapy, and targeted drugs, aggressive sub-types (such as HER2-positive and triple-negative cancers) or late-stage presentations carry poorer prognoses [21]. In many low- and middle-income regions—especially in Sub-Saharan Africa—limited access to diagnostic and treatment infrastructure leads to delayed presentation and high mortality [22]. This context underscores why effective detection is critical, and why certain populations face greater diagnostic challenges.

Breast cancer detection relies on a range of imaging and diagnostic methods, each with its own limitations. X-ray mammography remains the global gold standard because it can reveal masses and microcalcifications; however, its sensitivity drops sharply in women with dense breast tissue, falling to roughly 47.8–64.4% due to the masking effect [23]. Ultrasound, often used as a supplemental tool, offers improved sensitivity and specificity in dense breasts but is highly

operator-dependent, struggles with detecting microcalcifications, and can increase false positives [24]. MRI provides excellent tissue contrast and functional information, yet it is expensive, time-intensive, and associated with higher false-positive rates in some studies [25]. Clinical breast examination and self-examination may help identify palpable abnormalities but lack the sensitivity needed for reliable early detection and depend heavily on technique [26]. Biopsy methods—such as fine-needle aspiration, core-needle biopsy, and surgical biopsy—remain the diagnostic gold standard but are invasive, require pathology services, and cannot function as screening tools [27]. In low-resource settings, the financial and logistical barriers to mammography, ultrasound, MRI, and pathology services further contribute to late diagnosis and poorer outcomes [28].

Given these limitations, thermal imaging—also known as infrared thermography (IRT)—has gained renewed interest as a complementary detection tool [29]. IRT provides a functional, non-ionising approach by capturing surface temperature patterns linked to underlying physiological processes such as angiogenesis, metabolic activity, and increased blood flow around tumors [30]. A 2024 review of bio-heat transfer models highlights the potential of both steady-state and dynamic thermography to distinguish malignant from benign tissue when paired with advanced processing [31]. Recent studies report pooled sensitivities around 88.5% and specificities near 71.8%, although results vary due to differences in protocols and patient populations [32]. Emerging work using deep-learning techniques—achieving accuracies of up to 97% in classifying healthy, benign, and malignant breasts—further supports IRT’s potential [33], [34]. Although not a replacement for structural imaging, thermal imaging holds promise as an accessible, portable, non-invasive supplementary tool, particularly for women with dense breasts and in low-resource settings where conventional imaging methods remain limited [35].

CHAPTER THREE: METHODOLOGY

The methodology is divided into five major stages: (1) system requirements analysis, (2) hardware design and development, (3) software and algorithm development, (4) prototype integration, and (5) testing and evaluation.

A design and development research approach will be adopted. The project will combine principles of biomedical engineering, computer vision, and embedded systems to create a functional thermal screening prototype. The methodology incorporates both experimental design

(hardware testing, calibration, thermal measurements) and analytical techniques (image processing, algorithm development, and validation).

3.1 System Requirements Analysis

This stage involved identifying clinical, technical, and user-based requirements to guide device development.

3.1.1 Clinical Requirements

- Non-ionizing and safe for repeated use.
- Capable of detecting early physiological changes (thermal asymmetry, hot-spots)
- Suitable for low-resource environments in LMICs

3.1.2 Technical Requirements.

- Low-cost thermal sensing module.
- Real-time data acquisition and processing
- Reliable image reconstruction and noise suppression
- Portable power system
- User-friendly interface

3.1.3 User Requirements

- Minimal training required.
- Short screening time.
- Clear interpretation of outputs (normal vs suspicious)

The outcome of this analysis informed component selection and system architecture.

3.2 Hardware Design and Development

This phase focuses on selecting, assembling, and configuring components needed for thermal image acquisition.

3.2.1 Component Selection

Based on cost, resolution, and availability, the following components were selected:

- **Thermal imaging sensing module:** AMG8833 Thermal Image Sensor.
- **Processing unit:** NodeMCU ESP8833 WiFi Module
- **Power unit:** 3.7V Lithium-Ion Battery
- **Storage:** Micro SD card for data logging
- **Enclosure:** 3D-printed casing with thermal insulation

3.2.2 Sensor Integration

- The IR sensor will be interfaced with the NodeMCU ESP8833 WiFi Module through I²C/SPI.
- Calibration routines will be implemented to stabilize sensor readings.
- A fixed-distance mounting setup was added to ensure consistent imaging conditions.

3.2.3 Environmental Control

To reduce thermal interference:

- A stable imaging environment will be established (22–26°C).
- Subjects will be allowed a thermal relaxation period before imaging.

3.3 Software and Algorithm Development

3.3.1 Data Acquisition Module

Code will be developed to:

- Read sensor frames.
- Convert raw temperature data into 2D matrices.
- Apply basic filtering to remove noise.

Python Libraries used include numpy, opencv-python, adafruit-mlx90640 and matplotlib

3.3.2 Image Preprocessing

Preprocessing improves the quality and reliability of thermal images.

Steps include:

- Noise reduction using Gaussian filters.
- Contrast enhancement through histogram equalization.
- Hotspot stabilization.
- Temperature normalization

3.3.3 Region of Interest (ROI) Extraction

Algorithms will be developed to:

- Segment the breast region.
- Remove background objects.
- Align left and right breast regions for symmetry analysis

3.3.4 Thermal Feature Extraction

Several clinically relevant thermal features were computed:

- Maximum temperature difference (ΔT).
- Hot spot intensity and size.
- Temperature gradients.
- Vascular pattern signatures.
- Symmetry features between left and right breasts

3.3.5 Real-Time Classification Algorithm

A two-stage detection algorithm will be implemented:

1. Rule-based system:

Thresholding and asymmetry analysis identify suspicious areas.

2. Machine learning model (optional):

A lightweight CNN (TensorFlow Lite) classifies patterns as normal, benign, or potentially malignant.

The algorithm will output:

- **Normal**
- **Suspicious asymmetry**
- **High-risk thermal pattern**

3.4 User Interface Development

A graphical user interface (GUI) will be developed to:

- Display real-time thermal images.
- Present temperature maps and scales.
- Show detection results.
- Store patient screening data

The GUI will be implemented using either:

- **Tkinter** (lightweight option), or
- **PyQt5/PyQt6** (advanced interface option).

The interface will be designed for simplicity to allow use by non-specialized health workers.

3.5 Prototype Integration

All hardware and software components will be integrated into a compact, portable prototype.

3.5.1 Assembly

- Sensor and Pi will be mounted inside a heat-insulated casing.
- Battery and switches installed.
- Display screen connected (optional for portable screening)

3.5.2 System Validation

- Initial debugging to verify stable data flow.
- Testing of thermal image correctness.
- Verification of GUI functionality.

3.6 Testing and Evaluation

3.6.1 Laboratory Testing

Performed to assess:

- Sensor accuracy.
- Image stability.
- Calibration consistency.
- Ambient temperature effects.

Black-body references and controlled heat sources will be used for validation.

3.6.2 Algorithm Performance Evaluation

The system will be tested on:

- Thermal datasets (e.g., DMR-IR database).
- Simulated breast phantoms.

Metrics to evaluate will be:

- Sensitivity.
- Specificity.
- Accuracy.
- False-positive rate.
- Repeatability.

3.6.3 Usability Evaluation

User tests assessed:

- Interface clarity.
- Workflow simplicity.
- Screening time.
- Portability

Feedback informed refinements in UI and hardware.

3.7 Ethical Considerations

- All thermal imaging trials involving humans will require informed consent.
- No identifying anatomical images are stored without permission. • Thermal images are anonymized before analysis.
- The device will use **non-ionizing** infrared sensing, ensuring safety.

This methodology will ensure:

- Systematic development from requirements to testing.
- Compliance with clinical and engineering standards.
- A functional, low-cost, and portable device capable of early breast cancer thermal

screening.

- A research-backed approach to evaluating performance and feasibility

3.8 Work Plan

The project will run for six months, structured as shown below:

Task	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6	Milestone
Literature review & component selection	✓						Design finalized
Hardware procurement & setup		✓	✓				Functional assembly
Algorithm & software development		✓	✓	✓			Working prototype
Testing & data acquisition				✓	✓		Validated dataset
Analysis & performance evaluation					✓	✓	Accuracy report

Report writing& presentatio n					✓	✓	Final submiss io n
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3.6 Budget

Component	Example / Model	Estimated Cost (UGX)	Remarks
Microcontroller	NodeMCU ESP8833 WiFi Module	26,000	Controls sensor and processes data
Infrared Sensor	AMG8833 Thermal Image Sensor.	125,000	Medium-resoluti on, radiometric sensor
Display Unit	ILI9341 2.8" TFT LCD Display	50,400	Visual output
Power Supply	3.7V Lithium-Ion Battery.	19,000	Portable and rechargeable
	TP4056 Battery Charging Module.	29,000	

Auxiliary components	3-Pin Slide Switch 2kgs of Polylactic Acid (PLA)	-10,044 250,000	- For turning on/ off the device - For 3D Printing the
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			external casing
Software	Python,Arduino, Github	Free	Open-source tools
Miscellaneous	Cables, housing(3D Priented), screws, glue etc	15,000	Integration materials
Contingency	—	20,000	Covers variations

Total Estimated Cost: UGX 545, 000

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